CAS (Heimdall) – design basis for control

Parameters, dynamic model and linear model

Design basis version 2.3.5

Sigurd R. Brekk

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1 Purpose

This document sets out the parameters and the dynamic model used to design the level and oxygen control for the ScaleAQ closed aquaculture system (CAS), PLC project Heimdall. Every parameter has a value, a unit, a source and a reason for the choice. It replaces the parameter set in *CAS model.tex*; Section 9 lists the changes.

One source of truth. The parameters live in `design_basis.toml` [TOML] and the equations in `casdesign/design.py` [MODEL], both in `workfolder-python/CasDesignBasis`. Two simulators run them (Section 7): CasRioSim, the remote I/O over Modbus TCP and the pen physics, and CasFramoSim, the Framo inlet pump controllers over OPC UA. Their version is the design basis version (2.3.5). The tables, matrices and difference equations in this document are generated from the same files by `tools/design_report.py` [PY] into `generated/`. To change the design:

1. change `design_basis.toml` (or `design.py`) and bump `meta.version`;
2. run `python tools/selftest_plant.py`, which checks that the model linearizes to exactly the matrices in Section 6;
3. run `python tools/design_report.py` and rebuild this document;
4. push to `main`. CI publishes both simulator images with the new version.

Values marked [A] are assumptions that are not yet supported by a document. They are listed in Section 10 and must be confirmed.

Sources

ID	Document
[DB]	10925-CAS-001-02 Design basis, rev. 2, Automation (working document)
[FD]	P-15169-CAS-FD-01 Funksjonsbeskrivelse CAS, rev. B incl. tracked changes up to 28.09.26
[SPEC]	Framo 0000_CAS_SPEC_CTRLSYS, Kontrollsystem (prototype), rev. B, 08.10.2025
[SR2000]	Framo SR2000 pump specification (datasheet)
[PF]	pumpeflow_Sr2000_CAS.xlsx (pump power and system head reference)
[O2S]	Beregningsark dimensjonering ved bortfall av pumper eller strøm, 17.03.26
[CT]	Oppsett innløsere CT.xlsx (Chucaotech oxygenator sizing)
[CTE]	CT002-CS-022 v0-2, oxygen transfer efficiency test, 23.07.2024
[B2865]	Bürkert 2865 proportional valve, datasheet
[S418]	SUTO S418 flow meter, datasheet
[RDO]	RDO Core (Orbit 864) dissolved oxygen sensor, spec sheet
[VEGA]	VEGAPULS C 21 radar level sensor, specification sheet
[LTU]	LTU801 pressure level transmitter, IOM
[SO]	Sensor Overview CAS V1.xlsx
[B8605]	Bürkert 8605 PWM driver, Linde default setup, 22.02.22
[W455]	WAGO 750-455 4AI 4–20 mA product manual, 2453453963, valid from HW 07 / FW 04
[RDOM]	In-Situ RDO Core Digital Interface Specification (Modbus register map)
[W652]	WAGO 750-652 serial interface RS-232/RS-485 manual, version 2.0.0
[HEIM]	Heimdall CODESYS project (scaleaq/heimdall-plc): ModbusRTUData, GVL_CONSTANT
[TOML]	CasDesignBasis/design_basis.toml, version 2.3.5 (all parameters)
[MODEL]	CasDesignBasis/casdesign/design.py (model equations, used by the simulators)
[PY]	CasDesignBasis/tools/design_report.py (generates this document's tables)

2 System and control structure

Water is pumped into the pen by three Framo SR2000 inlet pumps through flexible tunnels taken in at 10–30 m depth. It leaves through a bottom centre outlet that has four hydraulic dampers. The pen level is held a few centimetres above the outside sea level (the *over-height* Δh), and that over-height drives the outflow. Oxygen is dissolved in a side stream on each inlet (oxy-pump and Chucaotech oxygenator), with one oxygen gas flow loop per line.

The control structure follows [FD] §3.2.2–3.2.4 (Figure 1):

- **LIC01**, level: PI on Δh_{avg}. It sends one common speed setpoint to all running inlet pumps. The dampers are held at a fixed opening per operating mode.

- **OIC01**, oxygen (master): PI on one pen O_2 sensor, normally OT03 at the outlet. Its output is the setpoint for the three O_2 flow loops.
- **FIC41–43**, O_2 flow (slave): PI on the gas flow (FT41–43). It drives the PWM proportional valve OV41–43.

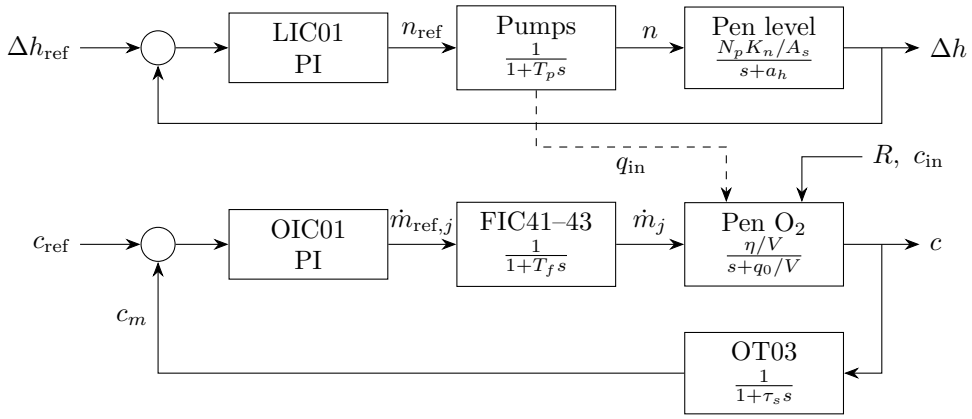


Figure 1: Control structure. The dashed line is the coupling from pump flow to oxygen.

3 Parameters

All tables in this section are generated from [TOML]; each row carries its own source and reason there.

3.1 Pen and outlet

Symbol	Value	Unit	Source	Reason
V	20 000	m^3	[DB] 1.2	Design volume.
D	50.5	m	[DB] tab. 7	Tarpaulin diameter. The free surface $A_s = \pi D^2/4$ sets the level capacitance; the surface is treated as rigid.
D_o	17.3	m	[DB] tab. 7	Tarpaulin depth, taken as the outlet depth.
ρ	1025	kg/m^3	[DB] tab. 21	Sea water density.
$\Delta\rho$	2	kg/m^3	[DB] 4.4.1	Density difference [DB] uses for the inner water height. Inside water from 10–30 m is assumed denser, giving the extra outlet head $h_\rho = D_o\Delta\rho/\rho$.[A]
full-open mode	O-45	–	[FD] tab. 10, [DB] tab. 16	Outlet conductance at full damper opening, taken from O-45. O-35 and O-45 give the same conductance (30.8 vs. 30.9), consistent with a fully open outlet. Conductance is assumed linear in the opening.[A]
A_s	2003	m^2	$\pi D^2/4$	Free surface area.
h_ρ	3.38	cm	$D_o\Delta\rho/\rho$	Extra driving head at the outlet.
$k_{d,\text{full}}$	30.94	$m^{2.5}/s$	$Q_0/\sqrt{\Delta h_0 + h_\rho}$	Outlet conductance at full opening (O-45).

3.2 Inlet pumps

Symbol	Value	Unit	Source	Reason
Q_{rated}	20 000	m^3/h	[SR2000]	Rated flow at rated speed.
$n_{\min}$	40	rpm	[SR2000], [FD] tab. 8	Minimum speed, also the LIC01 output range.
$n_{\max}$	130	rpm	[SR2000], [FD] tab. 8	Maximum speed. $K_n = Q_{\text{rated}}/n_{\max}$ by the affinity law $Q \propto n$, which holds because the system head is almost purely $\propto Q^2$ [PF] and the static over-height is below 10 cm.
N	3	–	[DB] 3.4	Three inlets, each with one SR2000 and one oxygenator.
T_p	10	s	-	Time constant from speed setpoint to actual speed (VSD ramp and water column). To be confirmed by Framo.[A]
$P_{\max}$	60	kW	[SR2000]	Motor power at $n_{\max}$. Power follows the affinity law $P = P_{\max}(n/n_{\max})^3$; at 130 rpm this gives the datasheet torque of 4407 Nm.
z_p	72	–	[SR2000]	Motor poles. Electrical frequency $f = nz_p/120$, 78 Hz at 130 rpm.
I_{rated}	110	A	[SR2000]	Rated current at 400 V. Current is scaled from power and voltage.
U_{rated}	400	V	[SR2000]	Rated voltage. Motor voltage is taken proportional to frequency (V/f).
K_n	0.042735	$(\text{m}^3/\text{s})/\text{rpm}$	$Q_{\text{rated}}/n_{\max}$	Flow per rpm.

3.3 Operating modes

Flow, level setpoint and number of pumps per mode come from [FD] tab. 10 and [DB] tab. 16. The other columns follow from the model: $n_0 = Q_0/(N_p K_n)$, $k_d = Q_0/\sqrt{\Delta h_0 + h_\rho}$, the damper opening $k_d/k_{d,\text{full}}$, and the level time constant $1/a_h$ (Section 6).

Mode	Q_0 [m^3/h]	Δh_0 [cm]	N_p	n_0 [rpm]	k_d [$\text{m}^{2.5}/\text{s}$]	Opening [%]	$1/a_h$ [s]
O-35	34 000	6.0	3	73.7	30.84	100	39.8
O-45	27 000	2.5	2	87.8	30.94	100	31.4
O-60	20 000	0.5	2	65.0	28.22	91	27.9
S-75	16 000	0.5	2	52.0	22.58	73	34.9

O-45 is the nominal design point. It is the [DB] dimensioning case (case 3: 1 000 t, 27 000 m^3/h,

45 min), and the mode that must be held with one pump out. The linear model and the discrete equations are given for O-45 with 2 pumps.

3.4 Water and oxygen

Symbol	Value	Unit	Source	Reason
T_w	10	°C	[DB] tab. 21	Design temperature.
S_w	34	ppt	[O2S]	Salinity.
$c_{\text{in}}/c_{\text{sat}}$	0.8	–	[O2S]	Intake water at 80 % O_2 saturation; saturation by the Weiss formula of [O2S].
$c_{\text{ref}}/c_{\text{sat}}$	0.9	–	[O2S]	Outlet design criterion, 90 % saturation. [FD] has no O_2 setpoint yet.[A]

Symbol	Value	Unit	Source	Reason
$\dot{m}_{\max}$	104.2	kg/h	[DB] tab. 13, [CT]	Maximum O_2 gas flow per oxygenator (CT10, sea water, 15 % gas/liquid ratio). An oxygenator adds O_2 only while its own inlet pump runs.
η	0.8	–	[CTE]	Transfer efficiency; the test gave 0.81 initial efficiency, rounded down.[A]
T_f	2	s	design choice	Closed-loop time constant of FIC41-43, used in the control design model.
τ_v	0.22	s	[B2865], [S418]	Open-loop lag of valve and flow meter, used in the simulator: valve < 25 ms, flow meter $t_{90} = 0.5$ s ($\tau_v = t_{90}/\ln 10$).
u_{cut}	2	%	[B8605]	Driver cut-off; below this the valve is closed (Linde default). Above it the gas flow is linear in the command up to $\dot{m}_{\max}$.
$u_{\%}$	4-20 mA	–	user, [B8605]	4 mA (0 %) = fully closed, 20 mA (100 %) = fully open. The PLC writes 0–65535 for 0–100 % to the 750-563 output (factory setting: 4–20 mA over the full 16-bit word); the Bürkert 8605 driver takes the 4–20 mA (Linde default) and sets the coil current. Below the cut-off the valve stays closed, so a lost signal is expected to close the valve (no O_2 on that line).
τ_s	19.54	s	[RDO]	O_2 sensor as first order fitted to $T_{90} < 45$ s ($\tau_s = T_{90}/\ln 10$). [RDO] gives $T_{63} < 5$ s, so this is conservative.
c_{sat}	9.09	g/m^3	[O2S]	Saturation at T_w, S_w (Weiss).
c_{in}	7.27	g/m^3		O_2 in the intake water.
c_{ref}	8.18	g/m^3		O_2 setpoint.
$\dot{m}_{\max}$	28.94	g/s		Maximum O_2 gas flow per line.

T_f is used in the control design model, where FIC41–43 is closed. τ_v and the cut-off are used in the simulator, where the PLC's FIC drives the valve itself (Section 7).

3.5 Fish

Symbol	Value	Unit	Source	Reason
ρ_f	50	kg/m^3	[DB] 1.2	Maximum density; biomass $B = \rho_f V$.
W	1000	g	[DB] tab. 18	Average weight, [DB] case 3.
U	41	cm/s	[O2S]	Recommended swimming speed for 1 kg at 10 °C.
B	1 000 000	kg	$\rho_f V$	Maximum biomass.
r	3.26	mg/(kg min)	[O2S]	Average post-smolt salmon, $r = 1.92\,W^{-0.27}T^{0.63}10^{0.01U}$ ($= 196$ mg/(kg h)).
R_0	54.3	g/s	rB	Nominal total O_2 consumption of the fish.

3.6 Level measurement

Symbol	Value	Unit	Source	Reason
LT01–03	2	mm	[VEGA], [SO]	Radar, deviation ≤ 2 mm.
LT21–23	4	mm	[LTU], [SO]	Pressure transmitter, 0.2 % of an assumed 2 m full scale.[A]
h_{span}	2	m	[VEGA], [W455], user	Span of the 4–20 mA level signal. The 750-455 puts the value in bits 14–4 of the word, 0–2047 (20 mA = 0x7FF0; bit 15 is the sign, bits 3–2 reserved, 1–0 diagnostics) [W455], confirmed by WAGO 2026-10-07. One step is 2000/2047 = 0.98 mm, inside the ±2 mm sensor deviation; the over-height, a difference of two readings, moves in steps of about 1 mm.
raw 0	low	–	[VEGA]	Which end of the span reads 0 (4 mA). "low": 0 = lowest water level in the span (VEGA level-proportional output, 4 mA = min. adjustment). "high": 0 = highest level, as when the output follows the measured distance from the antenna. Not stated in the datasheets; read the min./max. adjustment and current output mode in the sensor.[A]
$h_{\text{out},0}$	1	m	-	Outside sea level within the span, mid-range. Set by the sensor mounting height.[A]
OT01–03, OT21–22	5	–	[RDOM], [W652], [HEIM]	In-Situ RDO Core (Orbit 864), Modbus RTU on the RS-485 line of the 750-652, the first module on CommonRIO (Modbus words 0–11). Three inside the pen (slave addresses 1–3), two outside (4–5), as in Heimdall (GVL_CONSTANT); the cabinet drawing shows six. 9600 baud 8N1, as ScaleAQ configures them (Assembly Manual DOC100486-04). Idle sensors sleep and answer the waking request late; Heimdall retries, or writes Communication Stay Awake (register 9210). The inside sensors read c_m, the outside ones c_{in}. One sub-sensor

The PV range of LIC01 is 0–20 cm [FD] tab. 8, the mean over the active sensor pairs.

4 Dynamic model

4.1 States, inputs and disturbances

$x_1 = \Delta h$ [m], over-height
$x_2 = n$ [rpm], actual pump speed
$x_3 = c$ [g/m^3], O_2 concentration in pen
$x_4 = \dot{m}_1$ [g/s], O_2 gas flow line 1
$x_5 = \dot{m}_2$ [g/s], line 2
$x_6 = \dot{m}_3$ [g/s], line 3
$x_7 = c_m$ [g/m^3], measured O_2 (OT03)

$u_1 = n_{\text{ref}}$ [rpm], common pump speed setpoint
$u_2 = \dot{m}_{\text{ref},1}$ [g/s], O_2 flow setpoint line 1
$u_3 = \dot{m}_{\text{ref},2}$ [g/s], line 2
$u_4 = \dot{m}_{\text{ref},3}$ [g/s], line 3
$d_1 = R$ [g/s], O_2 consumption of fish
$d_2 = c_{\text{in}}$ [g/m^3], O_2 in intake water

The O_2 inputs are setpoints to the flow loops, not valve positions, because [FD] rev. B puts FIC41–43 in cascade under OIC01. To go from the [FD] valve scaling (0–100 %) to flow, use $\dot{m}_{\text{ref}} = \dot{m}_{\max} \cdot u_{\%}/100$.

4.2 Water balance

The inflow is the total pump flow. The outflow goes through the damper opening, driven by the over-height plus the density head:

$$q_{\text{in}} = N_p K_n \, n \tag{1}$$

$$q_{\text{out}} = k_d \sqrt{\Delta h + h_\rho} \tag{2}$$

$$A_s \, \dot{x}_1 = N_p K_n \, x_2 - k_d \sqrt{x_1 + h_\rho} \tag{3}$$

$$T_p \, \dot{x}_2 = -x_2 + u_1 \tag{4}$$

Here k_d is the outlet conductance at the fixed damper opening of the mode. It is set so that the mode's flow Q_0 gives the mode's over-height Δh_0: $k_d = Q_0/\sqrt{\Delta h_0 + h_\rho}$.

The outlet follows the orifice law ($q \propto \sqrt{h}$), not a linear resistance. That halves the slope compared with a linear outlet, and makes the level time constant twice as long.

4.3 Oxygen balance

The pen is treated as one fully mixed volume (CSTR). Since $\frac{d}{dt}(Vc) = q_{\text{in}} c_{\text{in}} - q_{\text{out}} c + \eta \sum \dot{m}_j - R$ and $\dot{V} = q_{\text{in}} - q_{\text{out}}$:

$$V \, \dot{x}_3 = N_p K_n \, x_2 \, (d_2 - x_3) + \eta \, (\sigma_1 x_4 + \sigma_2 x_5 + \sigma_3 x_6) - d_1 \tag{5}$$

$$T_f \, \dot{x}_{3+j} = -x_{3+j} + u_{1+j}, \qquad j = 1, 2, 3 \tag{6}$$

$$\tau_s \, \dot{x}_7 = -x_7 + x_3 \tag{7}$$

The oxygenators are additive. The intake water brings its own oxygen through the term $q_{\text{in}}c_{\text{in}}$, and each oxygenator adds $\eta\dot{m}_j$ on top of that. The side stream through each oxygenator (288–486 m^3/h [DB] tab. 13) is only 2–4 % of a pump's flow and goes back into the same inlet, so it does not appear in the water balance.

$\sigma_j \in \{0, 1\}$ says whether oxygen line j is in operation, and $N_o = \sum_j \sigma_j$ is the number of active lines. The oxygenator on a stopped inlet cannot add oxygen to the pen (confirmed; see also the interlock in [FD] tab. 2). Therefore $\sigma_j = 1$ only while inlet pump j runs, and $N_o = N_p$.

Equation (5) is exact for a fully mixed volume; no $\dot{V}$ term is lost. The water flow therefore acts on oxygen as $q_{\text{in}}(c_{\text{in}} - c)$. The intake water contains less oxygen than the setpoint ($c_{\text{in}} < c_{\text{ref}}$), so **more pump flow lowers the pen $\mathbf{O_2}$** while the fish consume oxygen and injection keeps c above c_{in}.

5 Operating points

At steady state $\Delta h = \Delta h_0$ and $c = c_{\text{ref}}$, and the N_o active lines share the load equally:

$$n_0 = \frac{Q_0}{N_p K_n}, \qquad \dot{m}_{j,0} = \frac{1}{N_o \eta}\Big(R_0 - Q_0\,(c_{\text{in}} - c_{\text{ref}})\Big) \tag{8}$$

Mode	V/Q_0 [min]	$N_p = N_o$	n_0 [rpm]	$\dot{m}_{j,0}$ [g/s]	Load of oxygenator [%]
O-35	35.3	3	73.7	26.22	90.6
O-45	44.4	3	58.5	25.48	88.0
O-45	44.4	2	87.8	38.22	**132.1**
O-60	60.0	3	43.3	24.75	85.5
O-60	60.0	2	65.0	37.12	**128.2**
S-75	75.0	2	52.0	36.49	**126.1**

The speeds in the table are inside 40–130 rpm. S-75 on three pumps would need 34.7 rpm, which is below the minimum speed, so S-75 can only run on two pumps. The oxygen requirement, not the water flow, decides how many pumps must run. At full biomass **three pumps must run in O-35, O-45 and O-60** (at reduced speed in O-45 and O-60). S-75 cannot hold c_{ref} at full biomass, and with two pumps $c = c_{\text{ref}}$ cannot be reached in any mode (Section 8). The linear model in the next section uses O-45 with two pumps, the redundancy case; around a 3-pump point only N_p and n_0 change.

6 Linear model

Linearize around the operating point (x_0, u_0, d_0) and write $\delta x = x - x_0$. Define

$$a_h = \frac{\partial q_{\text{out}}}{\partial \Delta h}\frac{1}{A_s} = \frac{Q_0}{2A_s(\Delta h_0 + h_\rho)}, \qquad a_c = \frac{Q_0}{V}, \qquad b_c = \frac{N_p K_n (c_{\text{in}} - c_{\text{ref}})}{V} \tag{9}$$

Then

$$\delta\dot{\mathbf{x}} = \mathbf{A}\,\delta\mathbf{x} + \mathbf{B}\,\delta\mathbf{u} + \mathbf{E}\,\delta\mathbf{d}, \qquad \mathbf{y} = \mathbf{C}\,\delta\mathbf{x} \tag{10}$$

$$\mathbf{A} = \begin{bmatrix} -a_h & \frac{N_p K_n}{A_s} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{T_p} & 0 & 0 & 0 & 0 & 0 \\ 0 & b_c & -a_c & \frac{\eta\sigma_1}{V} & \frac{\eta\sigma_2}{V} & \frac{\eta\sigma_3}{V} & 0 \\ 0 & 0 & 0 & -\frac{1}{T_f} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{T_f} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{T_f} & 0 \\ 0 & 0 & \frac{1}{\tau_s} & 0 & 0 & 0 & -\frac{1}{\tau_s} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ \frac{1}{T_p} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & \frac{\sigma_1}{T_f} & 0 & 0 \\ 0 & 0 & \frac{\sigma_2}{T_f} & 0 \\ 0 & 0 & 0 & \frac{\sigma_3}{T_f} \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{E} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ -\frac{1}{V} & a_c \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \tag{11}$$

The outputs are Δh_{avg}, the pump speed read back (MotorVelocity [SPEC]), OT03 and FT41–43:

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \tag{12}$$

Numerical values, O-45 with 2 pumps

Coefficient	Value	Unit
a_h	0.0319	1/s $\quad (1/a_h = 31.4\,\text{s})$
$N_p K_n / A_s$	4.2672×10^{-5}	m/(s rpm)
$1/T_p$	0.1000	1/s
a_c	3.7500×10^{-4}	1/s $\quad (1/a_c = 44.4\,\text{min})$
b_c	-3.8835×10^{-6}	g/(m^3 s rpm)
η / V	4.0000×10^{-5}	1/m^3 $\quad$ (times σ_j)
$1/T_f$	0.5000	1/s
$1/\tau_s$	0.0512	1/s
$1/V$	5.0000×10^{-5}	1/m^3

6.1 Transfer functions

The matrices are triangular, so the transfer functions are products of first-order terms:

$$\frac{\Delta h}{n_{\text{ref}}}(s) = \frac{N_p K_n / A_s}{(s + a_h)(1 + T_p s)} \qquad K = 1.339 \text{ mm/rpm} \tag{13}$$

$$\frac{c_m}{\dot{m}_{\text{ref},j}}(s) = \frac{\eta / V}{(s + a_c)(1 + T_f s)(1 + \tau_s s)} \qquad K = \frac{\eta}{Q_0} = 0.1067 \ \frac{\text{g/m}^3}{\text{g/s}} \tag{14}$$

$$\frac{c_m}{R}(s) = \frac{-1/V}{(s + a_c)(1 + \tau_s s)} \qquad K = -\frac{1}{Q_0} = -0.1333 \ \frac{\text{g/m}^3}{\text{g/s}} \tag{15}$$

$$\frac{c_m}{n_{\text{ref}}}(s) = \frac{b_c}{(s + a_c)(1 + T_p s)(1 + \tau_s s)} \qquad K = \frac{N_p K_n (c_{\text{in}} - c_{\text{ref}})}{Q_0} = -0.0104 \ \frac{\text{g/m}^3}{\text{rpm}} \tag{16}$$

$$\frac{c_m}{c_{\text{in}}}(s) = \frac{a_c}{(s + a_c)(1 + \tau_s s)} \qquad K = 1 \tag{17}$$

The steady-state gains are for O-45 and are computed by [PY].

6.2 Discretization

Each first-order term $\frac{1}{1+\tau s}$ discretizes exactly with zero-order hold (ZOH) to $\frac{1-p}{z-p}$, where $p = e^{-T_s/\tau}$. With $T_s = 1\,\text{s}$ at O-45:

Term	τ [s]	p
Level, $1/a_h$	31.4	0.968638
Pump speed, T_p	10.0	0.904837
Pen O_2, $1/a_c$	2666.7	0.999625
O_2 flow loop, T_f	2.0	0.606531
O_2 sensor, τ_s	19.5	0.950110

All the equations below are for O-45 with 2 pumps running and $T_s = 1\,\text{s}$. They are in deviation variables (the δ is dropped), and the inputs are held constant over each sample.

State update

The exact ZOH of the full state-space model is $\mathbf{x}_k = \mathbf{A}_d\mathbf{x}_{k-1} + \mathbf{B}_d\mathbf{u}_{k-1} + \mathbf{E}_d\mathbf{d}_{k-1}$, with $\mathbf{A}_d = e^{\mathbf{A}T_s}$. Row by row:

$$
\begin{aligned}
\Delta h_k &= 0.968638\,\Delta h_{k-1} + 3.995\,688\times10^{-5}\,n_{k-1} + 2.042\,285\times10^{-6}\,n_{\mathrm{ref},k-1}\\
n_k &= 0.904837\,n_{k-1} + 0.095163\,n_{\mathrm{ref},k-1}\\
c_k &= 0.999625\,c_{k-1} - 3.694\,972\times10^{-6}\,n_{k-1} + 3.147\,116\times10^{-5}\,(\dot m_{1,k-1} + \dot m_{2,k-1})\\
&\quad - 1.878\,394\times10^{-7}\,n_{\mathrm{ref},k-1} + 8.521\,345\times10^{-6}\,(\dot m_{\mathrm{ref},1,k-1} + \dot m_{\mathrm{ref},2,k-1})\\
&\quad - 4.999\,063\times10^{-5}\,R_{k-1} + 3.749\,297\times10^{-4}\,c_{\mathrm{in},k-1}\\
\dot m_{j,k} &= 0.606531\,\dot m_{j,k-1} + 0.393469\,\dot m_{\mathrm{ref},j,k-1}, \qquad \text{lines with } \sigma_j = 1\\
c_{m,k} &= 0.950110\,c_{m,k-1} - 9.449\,822\times10^{-8}\,n_{k-1} + 0.049880\,c_{k-1}\\
&\quad + 8.569\,178\times10^{-7}\,(\dot m_{1,k-1} + \dot m_{2,k-1}) - 3.189\,862\times10^{-9}\,n_{\mathrm{ref},k-1}\\
&\quad + 1.492\,579\times10^{-7}\,(\dot m_{\mathrm{ref},1,k-1} + \dot m_{\mathrm{ref},2,k-1}) - 1.257\,720\times10^{-6}\,R_{k-1}\\
&\quad + 9.432\,897\times10^{-6}\,c_{\mathrm{in},k-1}
\end{aligned}
$$

Lines with $\sigma_j = 0$ do not appear in the c equations.

Input–output

Each transfer function from Section 6.1 is discretized exactly with ZOH.

$$\frac{Y_1(z)}{U_1(z)} = \frac{\Delta h}{n_{\mathrm{ref}}} = \frac{2.042\,285\,28\times10^{-6}\,z + 1.954\,463\,26\times10^{-6}}{z^2 - 1.87347534\,z + 0.87645984} \tag{18}$$

$$\frac{Y_3(z)}{U_2(z)} = \frac{c_m}{\dot m_{\mathrm{ref},j}} = \frac{1.492\,578\,58\times10^{-7}\,z^2 + 5.224\,854\,70\times10^{-7}\,z + 1.133\,103\,18\times10^{-7}}{z^3 - 2.55626615\,z^2 + 2.13232854\,z - 0.57605504} \tag{19}$$

$$\frac{Y_3(z)}{D_1(z)} = \frac{c_m}{R} = \frac{-1.257\,719\,57\times10^{-6}\,z - 1.236\,291\,92\times10^{-6}}{z^2 - 1.94973549\,z + 0.94975419} \tag{20}$$

$$\frac{Y_3(z)}{U_1(z)} = \frac{c_m}{n_{\mathrm{ref}}} = \frac{-3.189\,861\,71\times10^{-9}\,z^2 - 1.228\,717\,97\times10^{-8}\,z - 2.957\,079\,36\times10^{-9}}{z^3 - 2.85457291\,z^2 + 2.71394782\,z - 0.85937313} \tag{21}$$

$$\frac{C(z)}{U_2(z)} = \frac{c}{\dot m_{\mathrm{ref},j}} = \frac{8.521\,344\,72\times10^{-6}\,z + 7.214\,478\,24\times10^{-6}}{z^2 - 1.60615573\,z + 0.60630325} \tag{22}$$

As difference equations:

$$
\begin{aligned}
y_{1,k} &= 1.87347534\, y_{1,k-1} - 0.87645984\, y_{1,k-2} + 2.042\,285\,28 \times 10^{-6}\, u_{1,k-1} \\
&\quad + 1.954\,463\,26 \times 10^{-6}\, u_{1,k-2} \\
y_{3,k} &= 2.55626615\, y_{3,k-1} - 2.13232854\, y_{3,k-2} + 0.57605504\, y_{3,k-3} \\
&\quad + 1.492\,578\,58 \times 10^{-7}\, u_{2,k-1} + 5.224\,854\,70 \times 10^{-7}\, u_{2,k-2} + 1.133\,103\,18 \times 10^{-7}\, u_{2,k-3} \\
y_{3,k} &= 1.94973549\, y_{3,k-1} - 0.94975419\, y_{3,k-2} - 1.257\,719\,57 \times 10^{-6}\, d_{1,k-1} \\
&\quad - 1.236\,291\,92 \times 10^{-6}\, d_{1,k-2} \\
y_{3,k} &= 2.85457291\, y_{3,k-1} - 2.71394782\, y_{3,k-2} + 0.85937313\, y_{3,k-3} \\
&\quad - 3.189\,861\,71 \times 10^{-9}\, u_{1,k-1} - 1.228\,717\,97 \times 10^{-8}\, u_{1,k-2} - 2.957\,079\,36 \times 10^{-9}\, u_{1,k-3} \\
c_k &= 1.60615573\, c_{k-1} - 0.60630325\, c_{k-2} + 8.521\,344\,72 \times 10^{-6}\, u_{2,k-1} \\
&\quad + 7.214\,478\,24 \times 10^{-6}\, u_{2,k-2}
\end{aligned}
$$

Check with the final value theorem, $y[\infty] = \lim_{z\to 1}(z-1)\,G(z)\frac{z}{z-1} = G(1)$: the steady-state gains are 1.339 mm/rpm, 0.1067, -0.1333 and -0.0104 for the four paths, the same as in continuous time.

The contributions from each input add up. For example, the full measured O_2 is $y_{3,k}$ = (response to u_2) + (response to u_3) + (response to d_1) + (response to u_1), each computed with its own equation above.

7 Simulators

Two simulators run this design basis, split along the PLC's real interfaces:

- **CasRioSim**: the four Wago remote I/O stations over Modbus TCP, and the pen physics (level, oxygen, valves, outlet, sensors);
- **CasFramoSim**: the three Framo inlet pump controllers over OPC UA (Section 7.2).

They share nothing but OPC UA and the design basis: CasRioSim's physics reads each pump's actual speed from its CasFramoSim server, as an OPC UA client. Both report the design basis version, 2.3.5.

7.1 Remote I/O and pen physics (CasRioSim)

CasRioSim serves the Heimdall PLC the same Modbus TCP process image as the four Wago remote I/O stations. The plant behind it is this design basis: it integrates the nonlinear equations (3)–(7) from [MODEL] with the parameters in [TOML], using RK4 at 0.1 s. The simulator version is the design basis version, 2.3.5; every station reports it in holding registers 2013–2015.

The simulator differs from the control design model only where the PLC, not the plant, owns the behaviour:

- **One speed per pump.** Each inlet has its own speed state. With CasFramoSim running, the speed is read from the pump's Framo server, which runs the lag $T_p\dot{n}_j = n_{\mathrm{ref},j} - n_j$ itself;

otherwise CasRioSim integrates it from a setpoint in its control block. With a common speed this is equation (4).

- **O_2 valve instead of the closed FIC.** The PLC runs FIC41–43 and writes the valve position $u_{\%,j}$. The plant is the valve itself: $\tau_v\,\ddot{m}_j = \sigma_j\,\dot{m}_{\max}\,u_{\%,j}/100 - \dot{m}_j$, zero below the cut-off u_{cut}. FT41–43 read $\dot{m}_j$.

- **Outlet opening.** $q_{\text{out}} = k_{d,\text{full}}\,z\,\text{sgn}(\Delta h + h_\rho)\sqrt{|\Delta h + h_\rho|}$ with opening $z \in [0,1]$, set from the simulator control block or from the mean damper cylinder position. A negative q_{out} is sea water flowing in through the outlet; it carries c_{in}.

- **Start state.** Pumps stopped, $\Delta h = -h_\rho$, $c = c_{\text{in}}$.

The self-test (`CasDesignBasis/tools/selftest_plant.py`) checks that every operating point in Section 5 is an equilibrium of the simulator plant, and that the plant linearizes to exactly **A**, **B** and **E** of Section 6 in every mode.

The cylinder counters, CO_2, total gas pressure and oxy-pump pressures are simulated instruments with static curves; they are not part of the design model.

O_2 sensors. The PLC reads the Orbit 864 sensors (In-Situ RDO Core) over Modbus RTU through the 750-652 serial module, the first module on CommonRIO, so its 24-byte process image is Modbus words 0–11 [W652]. CasRioSim runs that module's handshake on the words the PLC writes and reads with FC23 (initialization IR/IA, transmit TR/TA with length OL, receive RR/RA with length IL, up to 22 bytes per step), and behind it five Modbus RTU slaves: OT01–03 inside the pen (addresses 1–3) read c_m, OT21–22 outside (4–5) read c_{in} [HEIM]. Each answers function code 3 with the register map of a real Orbit 864, read from the sensor on the test rig on 7 October 2026. The sensor is an RDO Core IC: one sub-sensor per register block, each laid out like the oxygen block of [RDOM] (sensor id at index 0, parameters from index 37, 8 registers apart):

- oxygen, registers 40001–40066, sensor id 42: DO in mg/L as a float at 40038–39 (parameter Id 20, unit 117), temperature at 40046, saturation at 40054 (from the same saturation formula as c_{sat}) and O_2 partial pressure at 40062;

- conductivity, 40873–40957, sensor id 85: actual and specific conductivity, salinity, total dissolved solids, resistivity and water density at 40910, 40918, 40926, 40934, 40942 and 40950;

- pressure, 41745–41797, sensor id 84: pressure (psi) at 41782 and depth (m) at 41790.

[RDOM] documents only the oxygen block. Register 4xxxx is PDU address xxxx−1, and floats are sent high word first; both were confirmed with consistent readings (8.68 mg/L at 24.4 °C is 103.8 % saturation). The "PLC interface" of [RDOM] (register $5451 + 7(\text{id} - 1)$) is refused by the sensor. The reply reaches the PLC after its wire time at 9600 baud 8N1 (as ScaleAQ configures the sensors). A sensor left idle sleeps and answers the request that wakes it about 3 s late, as seen on the test rig; register 9210 (Communication Stay Awake) keeps it awake. One poll of the oxygen block takes about nine Modbus TCP round trips, some 0.6 s. The module's register communication is not documented; the simulator answers it generically. The same value is in the simulator control block (holding register 2008, mg/L×100) for use without a PLC.

7.2 Framo inlet pump controllers (CasFramoSim)

Each inlet pump has its own Framo controller with an OPC UA server [SPEC]. CasFramoSim runs three identical servers, one container each, with the address space Heimdall's Data Source Manager binds to: the same names, numeric node ids and data types, in namespace `FramoPLS_1`–`3`. The node list is generated from the Data Source object (`framo_nodes.csv`, 305 nodes per pump); the ports are set per container (default 4841–4843, as in the Data Source endpoints).

Signal	Node (value)	Type	PLC
SpeedSetpoint	i=33 (`.Value` i=261)	AnalogValue [rpm]	writes
Start, Stop, Reset	i=30–32 (`.Active` 257–259)	Event	writes
HeartbeatToggle	i=34 (`.Active` i=277)	BOOL	writes
MotorVelocity	i=2 (`.Value` i=59)	AnalogValue [rpm]	reads
MotorFrequency, Current, Power, Voltage, Torque	i=1, 3–6	AnalogValue	reads
PumpState… (7 states)	i=23–29 (`.Active` 250–256)	Event	reads
Alarms (11)	i=12–22 (`.Active`, `.Id`)	Alarm	reads
Simulation (fault injection)	i=40 (members 299–305)	BOOL / REAL	test only

The controller follows the state table in [SPEC] p. 6: UNAVAILABLE → AVAILABLE when no fault is active; Start (rising edge) → START, where the controller ramps the pump to n_{start}; at $n_{\text{start}} \pm \Delta n_{\text{start}}$ ("Sequence END") → REMOTE, where the pump follows SpeedSetpoint clamped to $n_{\min}$–$n_{\max}$. Stop → STOP, which ramps to zero and returns to AVAILABLE. A fault (DriveError, MotorTemperature, EmergencyStop, UPSSupplyError, DCUnderVoltageError) in START, REMOTE, HOLD or STOP → LOCKDOWN, which ramps to zero and ends in UNAVAILABLE. With heartbeat supervision on, REMOTE → HOLD (start speed) when HeartbeatToggle stops changing, and back to REMOTE when it toggles again. The ramp is the plant's pump lag T_p, run in CasFramoSim; CasRioSim reads the resulting speed over OPC UA, so the simulated level follows the PLC's speed setpoint.

Drive feedback follows the SR2000 data: $f = nz_p/120$, $P = P_{\max}(n/n_{\max})^3$, torque P/ω (4407 Nm at 130 rpm, as on the datasheet), voltage $\propto f$, current scaled from power and voltage to 110 A at full speed. The evacuation (vacuum) nodes are present but static.

Symbol	Value	Unit	Source	Reason
n_{start}	88	rpm	[FD] 3.2.1	Speed the Framo controller ramps to after Start, before handing over in REMOTE ([FD]: "forhåndsinnstilt hastighet"). Set near n_0 in the nominal mode so the level is close to its setpoint at hand-over.[A]
Δn_{start}	1	rpm	-	START ends ("Sequence END") when the actual speed is within this of n_{start}; STOP and LOCKDOWN end below it.[A]
HB	off	–	[SPEC] p. 6	If true, REMOTE goes to HOLD when HeartbeatToggle has not changed for the timeout, and back to REMOTE when it toggles again. Off by default: Heimdall does not send the heartbeat yet.
T_{HB}	5	s	-	Heartbeat supervision timeout. [SPEC] has a HeartbeatInterval signal, but the Data Source Manager does not map it.[A]
$T_{\text{mot,trip}}$	135	°C	[SR2000]	Simulation.MotorTemperature above this raises the MotorTemperature alarm. Class B rise of 80 K over ambient, below the class F limit of 155 °C.[A]

8 Design checks

8.1 Oxygen capacity

These are the loads on each oxygenator needed to hold c_{ref} in O-45, and the largest biomass the running lines can carry, for the cases in [TOML]:

		3 pumps		2 pumps	
Case	R [g/s]	Load [%]	Max. biomass	Load [%]	Max. biomass
10 °C, 1.0 kg (nominal)	54.3	88.0	1153 t	**132.1**	727 t
12 °C, 1.0 kg ([DB] dimensioning)	61.0	97.1	1032 t	**145.7**	653 t
16 °C, 1.5 kg	71.8	**112.0**	884 t	**168.0**	561 t

This is the most important finding. Each oxygenator delivers only while its own inlet pump runs. Consequences:

- **Normal operation:** 1 000 t is feasible only with all three pumps running, and with little margin: 3 % at 12 °C, and not enough at 16 °C/1.5 kg.
- **Redundancy case (one pump out):** [DB] covers the water flow with two pumps, but not the oxygen. Two oxygenators carry only about 560–730 t. Losing a pump at full biomass is therefore an oxygen event, not only a flow event; alarms and fallback (emergency oxygenation, stopping feeding) must be designed for it.

To resolve this, do one of the following:

- size each oxygenator for $\approx 1.5\times$ the present load (about 150–175 kg/h), so that two lines carry full biomass;
- limit the biomass to what two lines can carry, about 650 t (33 kg/m^3) at 12 °C;
- accept the shortfall for the duration of a pump outage, with emergency oxygenation ([FD] §3.2.5) and stopped feeding (2.5 mg/(kg min) [O2S]) as the defined fallback.

8.2 Level loop

- The level time constant is 28–40 s with h_ρ, and 4–25 s without it. Whether h_ρ is included is the largest single uncertainty in the level model.
- With h_ρ, the outlet conductance k_d is almost the same in O-35 and O-45 (30.8 vs. 30.9). The [DB] level setpoints are thus consistent with one fixed opening plus the density head, which supports the assumption.
- The process gain grows with N_p. The controller gain should therefore fall as more pumps run ($K_c \propto 1/N_p$). The [FD] table has it the other way round (0.5/1/2 for 1/2/3 pumps).
- In [FD] scaling (PV 0–20 cm, OUT 40–130 rpm) the normalized gain at O-45 is 0.60. With $T_i = 600$ s, which is about $20\times$ the process time constant, the closed loop becomes very slow. As a starting point, SIMC with $\tau_c = 30$ s gives $K_c \approx 1.3$ and $T_i \approx 31$ s. Verify by simulation.
- The setpoint in O-60/S-75 is 5 mm, and the measurement uncertainty is about ± 6 mm (± 2 mm inside, ± 4 mm outside). With waves up to $H_s = 2$ m, this calls for heavy filtering, so the level loop cannot be fast.

8.3 Oxygen loop

- The pen behaves almost as an integrator (time constant 44 min) behind $\tau_s + T_f \approx 22$ s of lag, plus an unknown mixing delay θ.
- The [FD] FIC parameters ($K_c = 0.5$, $T_i = 600$ s) make the inner loop slower than the outer one, which defeats the cascade. The inner loop should have T_i of the order of 1 s.
- The coupling from n is negative and small, $-0.010\,(\mathrm{g/m^3})/\mathrm{rpm}$. A step of 10 rpm in the level loop moves O_2 by about -0.1 g/m^3. Feedforward from pump flow to $\dot{m}_{\mathrm{ref}}$ cancels it: $\Delta\dot{m} = -\frac{N_p K_n (c_{\mathrm{in}} - c_{\mathrm{ref}})}{\eta}\Delta n$.
- A feedforward base dose $\dot{m}_{\mathrm{ff}} = \frac{1}{N_o \eta}(rB - q_{\mathrm{in}}(c_{\mathrm{in}} - c_{\mathrm{ref}}))$ is what the [FD] review comment asks for (base oxygenation from fish count, size and temperature). It also gives a safe fallback if the O_2 sensor is lost.

9 Changes from CAS model.tex

Item	Old	New	Why
Capacitance C	1288 / 1886 m^2	2003 m^2	$D = 50.5\,\mathrm{m}$ [DB]. The old values match $r = 20.25$ and $r = 24.5\,\mathrm{m}$.
Outlet	linear, $K_{qo}/0.06$	$k_d\sqrt{\Delta h + h_\rho}$	Orifice law; k_d per mode from [FD]/[DB].
Outflow	fixed 9.52 m^3/s	per mode, 4.4–9.4 m^3/s	Mode table [FD] tab. 10.
O_2 washout	$K_{qo}x_2/V$	$q_{\mathrm{in}}(c_{\mathrm{in}} - c)$	Exact mixing balance. The old model counted the intake oxygen both as K_2u and through a fixed washout.
O_2 in intake	7 g/m^3 (100 % inflow)	7.27 g/m^3 (80 % sat.)	[O2S].
Fish consumption	500 mg/(kg h), 139 g/s	196 mg/(kg h), 54.3 g/s	Average post-smolt formula of [O2S] at 10 °C, 1 kg.
Valve gain	2.9 / 0.29 / 28.9	flow setpoint in g/s	Cascade [FD] rev. B; the valve sits inside FIC.
Valve time constant	0.02 / 2 s mixed	$T_f = 2\,\mathrm{s}$ (closed FIC)	One consistent value.
Transfer efficiency	1.0	0.8	[CTE].
O_2 sensor	not modelled	$\tau_s = 19.5\,\mathrm{s}$	[RDO].
Pump dynamics	none	$T_p = 10\,\mathrm{s}$[A]	To be confirmed by Framo.

10 Assumptions and open items

1. h_ρ **(density head).** This assumes the water inside, taken from 10–30 m, is 2 kg/m^3 denser than the surface water outside. Confirm against how [DB] computes the inner water height. Set $h_\rho = 0$ if the densities are equal.

2. T_p**.** Ramp and response of the Framo VSD and the water column, from setpoint to speed. Ask Framo [SPEC].

3. **O_2 setpoint.** [FD] does not yet give a setpoint, nor whether the unit is % or mg/L. 90 % saturation is taken from [O2S].

4. **Mixing.** A fully mixed pen is assumed. The delay from injection to OT03 is not known; it needs CFD ([DB] §4.4.3) or a step test at commissioning.

5. η**.** The value 0.8 comes from a small-scale test (900 l, CT1). Confirm for CT10 at full scale.

6. **FS of LT21–23.** A full scale of 2 m is assumed for the accuracy.

7. **Direction of the level signal.** The level span is 2 m, which the 750-455 resolves in 2048 steps (0.98 mm per step). The VEGAPULS C 21 measures the distance from its antenna down to the water [VEGA]. Whether raw 0 (4 mA) is the lowest or the highest level depends on the min./max. adjustment and current output mode set in each sensor; the datasheets do not say. Read the settings in the sensors (VEGA Tools app). Inside (LT01–03) and outside (LT21–23) must have the same direction, otherwise the over-height, and the sign of LIC01, is wrong. The simulator assumes 0 = lowest level.

8. **Analog input word layout** (settled). The 750-455 puts the value in bits 14–4 (0–2047, 20 mA = `0x7FF0`); bit 15 is the sign, always 0 for 4–20 mA, and WAGO counts it in the stated 12-bit resolution. Bits 3–2 are reserved and bits 1–0 are diagnostics [W455]. WAGO Germany confirmed this layout on 7 October 2026 and called older tables (value in bits 14–3, 20 mA = `0x7FF8`) inconsistent. On a 2 m span one step is 0.98 mm, so the over-height, the difference of two readings, moves in steps of about 1 mm against level setpoints of 5 mm in O-60 and S-75. The simulator serves this layout. Heimdall's `RawMax` of 32760 reads 20 mA as 99.98 % of span.

9. **Quality check in the PLC.** The diagnostic bits are 0 on a healthy channel and set on a fault (both diagnostic bits are 1 on wire break or out of range [W455]). Heimdall's `AnalogSensor.ScaleInput` masks bits 0–2 as "quality" and accepts the value only when they are *non-zero*, so a healthy channel is reported as bad quality and a wire break is accepted as good. Suggested: fault when `(word AND 3) = 3`.

10. **Over-height in raw words.** Heimdall's `deltaHeightCalc` subtracts the inside and outside raw WORDs directly, without scaling to metres. Status bits enter the difference, and a WORD difference wraps when the inside level is below the outside one (e.g. $\Delta h = -h_\rho$ at rest). Computing the over-height from the scaled sensor values avoids both.

11. **Framo commands not sent.** In Heimdall, `InletPump.InternalCommand` (Start, Stop, Reset, VelocitySetpoint) is only read, in `FramoDataProcessing`, and never written, so no command or speed setpoint reaches the pumps yet.

12. **Framo heartbeat not wired.** `InletControl` supervises a local `HeartbeatToggle` that nothing changes, and `FramoDataProcessing` only reads `Framo_n.HeartbeatToggle`. A Framo controller that follows [SPEC] will go REMOTE → HOLD. The simulator's supervision is off until this is wired.

13. **Copy-paste errors in the Framo data processing.** In `FramoDataProcessing`, the inlet 2 and 3 blocks write Framo_1's `PressureCompressedAir` into `Inlet[0]`, and Framo_3's state mapping writes `Inlet[1].State` in every branch except Unavailable.

14. **Framo interface vs. [SPEC].** The Data Source names differ from [SPEC] (SpeedSetpoint vs. VelocitySetpoint, DCUnderVoltageError vs. DCUndervoltageError, PowerConsumption-Counter vs. PowerConsumtionCounter); HeartbeatInterval and HeartbeatElapsed are not mapped; Limit*, DisplayName and Description are BOOL instead of REAL/STRING; each pump has its own namespace URI. Confirm against the delivered Framo PLCs. The simulator follows the Data Source.

15. **Speed setpoint at hand-over.** In REMOTE the pump follows SpeedSetpoint at once. If the PLC has not written it (0), the pump drops to $n_{\min}$. The PLC should track the actual speed into SpeedSetpoint during START ([FD] bumpless transfer).

16. **O_2 valve output scaling** (settled). The 750-563 in its factory setting (register 32 = `0x8016`) outputs 4–20 mA over the full 16-bit word: `0x0000` = 4 mA (closed), `0x7FFF` = 12 mA, `0xFFFF` = 20 mA (fully open). Heimdall's `AnalogValve.ScaleOutput` writes (Setpoint − Min)/(Max − Min) · 65535, limited to 0–65535, and the simulator decodes that range. After 100 ms without process data the module outputs 0 mA, below the cut-off, so the valve closes.

17. **Rigid surface.** The tarpaulin is flexible, so the real capacitance may be larger.

18. **Maximum flow.** [DB] gives 27 000 m^3/h as the maximum, while O-35 needs 34 000 m^3/h [FD]. The two documents must be aligned.

19. **Valve capacity.** A rough estimate puts one Bürkert 2865 (4 mm orifice) at about 57 kg/h of O_2. The nominal need is 92 kg/h per line (25.5 g/s), so check the valve against $\dot{m}_{\max}$. The Linde default setting of the 8605 driver (maximum 860 mA) is also above the 750 mA coil limit [B2865].

20. **Structure in the PLC function flow diagram.** The diagram drives the valves directly from OIC01, with no FIC. [FD] rev. B has the cascade, which this model follows. Align the diagram with [FD].

21. **O_2 sensor update rate.** The Orbit 864 delivers a value every 2–4 s, so T_s for OIC01 should be no shorter than that.